Experiment–1

NAME: Tanisha.S **REGISTER NO: 240701553**

Design a UI where users recall visual elements (icons/text chunks) and evaluate the effect of chunking on user memory.

FRAME 1:

INSTRUCTION PAGE:

Chunking Analysis of the Instruction Page

Chunking is a cognitive strategy that breaks down information into smaller manageable units. The Memory Recall Task instruction page effectively utilizes chunking in the following ways:

1. Clear and Sequential Numbering

- The instructions are broken into six steps, making it easier to follow.
- Each step presents one key action, reducing cognitive overload.

2. Logical Grouping of Information

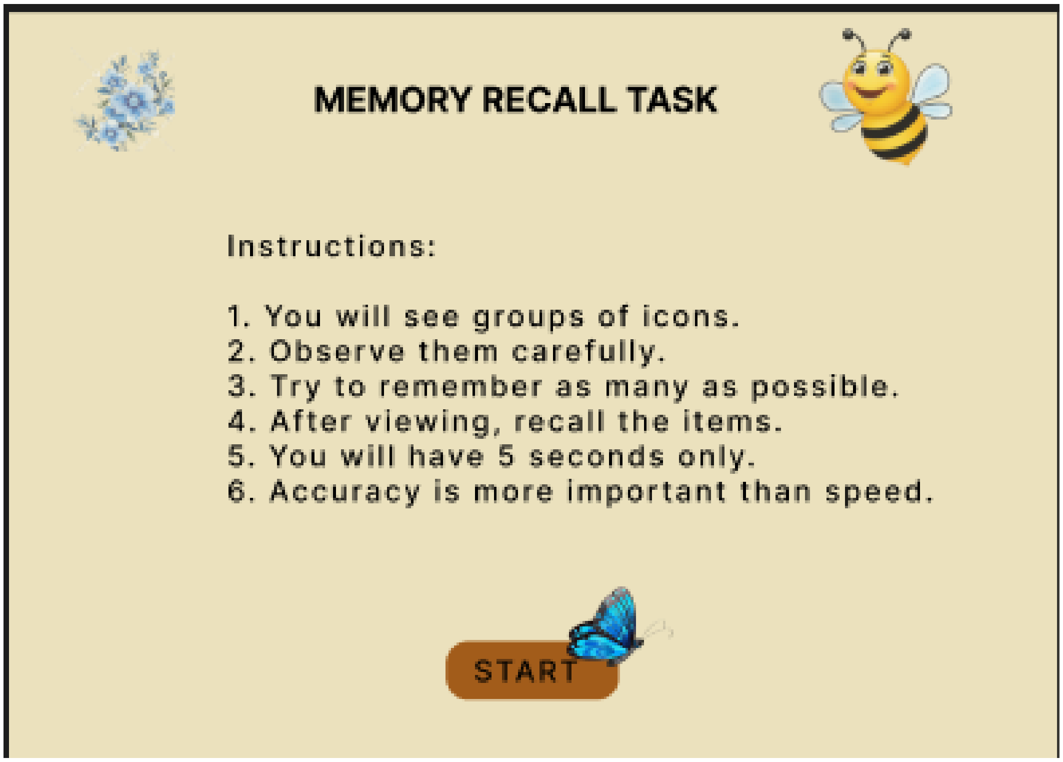
- **Observation Phase** – Users understand what they will see.
- **Memorization Strategy** – Users focus on remembering items.
- **Recall Phase** – Users recall information.
- Accuracy emphasis highlights correctness over speed.

3. Visual Hierarchy and Design Elements

- Bold title grabs attention.
- Bullet points improve readability.
- **START** button signals next step clearly.

4. Simplicity and Clarity

- Short direct sentences are used.
- Easy language avoids confusion.



FRAME 2:

CHUNKING PHASE:

Analysis of the Memory Recall Task – Chunking Phase Screen

1. Purpose of the Screen

- This is the **visual memory encoding phase**.
- Users observe and group items mentally before recalling.

2. Key Elements and UI Components

- Countdown timer shows **5 seconds.**
- **Progress bar** indicates remaining time.
- Grid of icons displayed in **5x4 format.**
- Icons grouped by theme (food, animals, vehicles, emojis).

3. How the Chunking Phase Works

- Users scan grid and identify patterns.
- **Timer limits** observation time.
- System transitions automatically to recall phase.

4. Cognitive & UX Benefits of Chunking

- Improves short-term memory.
- Reduces cognitive load.
- Enhances pattern recognition.

FRAME 3:
RECALL PHASE:

Analysis of the Memory Recall Task – Selection Phase

1. Purpose of the Screen

- This is the **memory retrieval** stage.
- Users select items they remember.

2. Key Elements and UI Components

- Instruction heading guides user.
- Grid of icon choices displayed.
- **Distractors** included for accuracy testing.
- **Radio buttons** enable selection.
- **Submit** button confirms answers.

3. How the Selection Phase Works

- Users analyze icons.
- Select remembered ones.
- Submit to view results.

4. Cognitive & UX Benefits

- Tests recall accuracy.
- Improves engagement.

- **Gamified** interface keeps users motivated.

FRAME 4:

RESULT PAGE:

Analysis of the Memory Recall Task – Score & Feedback Screen

1. Purpose of the Screen

- Displays feedback on recall accuracy.
- **Shows score** clearly.

2. Key Elements & UI Components

- Score board display.
- **Continue**, **Restart, Exit** buttons.

3. How This Phase Works

- System calculates score.
- Displays **accuracy**.
- User chooses next action.

4. Cognitive & UX Benefits

- Immediate feedback.
- Encourages improvement.
- Reduces stress through **gamification.**

CONCLUSION:

The experiment demonstrates that chunking significantly improves memory recall and reduces cognitive load. Grouping related items helps users remember information faster and more accurately compared to random placement. Therefore, chunking is an effective strategy for designing memory-based interfaces.